

APPLICATION OF DEPTH CAMERA TECHNOLOGY FOR EVIDENCE MEASUREMENT IN PRESUMPTIVE FORENSIC DNA LABORATORY

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Abstract

The precision and uncontaminated metric evaluation of evidence at crime scenes is essential for forensic investigations, especially for DNA examinations. The use of Time-of-Flight (ToF) depth cameras represents a significant advance in the field of forensic measurement, addressing limitations of existing metrology techniques. A top-down camera system was utilized to capture point cloud data corresponding to frequently found evidence in crime scenes, such as bottles, glass, knives, scissors, firearms, and other tools. An automated algorithm was developed to process the point cloud by isolating the evidence from the substrate and determining its principal dimensions: length, width, and depth. Our analysis depends on the resolution of the depth camera. However, we exclusively utilize the metrics of length and width for presenting the result of the evidence measuring report. A validation study was carried out to examine the value of depth cameras compared to measurements obtained manually and two-dimensional imaging technology. The data suggest that the depth camera equipment reaches a higher level of accuracy, with relative errors for both length and width measurements observed at 0.23. In addition, it adeptly addresses planar and perspective distortions that are conventionally associated with two-dimensional imaging techniques. This methodological approach demonstrates a significant advance in the presumptive investigation of forensic DNA examination processes, enabling accurate documentation of evidence. This enhances the fidelity of the forensic workflow and implies improved efficiency within the presumptive process, saving time and improving accuracy in metrology. Its usability contributes to the prevention of presumptive evidence contamination in the measuring phase.

Keywords: Depth Camera Measurement; Forensic Science; 3D Metrology; Point Cloud

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Introduction

Presumptive evidence in the forensic DNA laboratory is an important step before examination by machine. These tests consist of physical assessments that utilize photographic and measurement dimensions to examine forensic DNA evidence. For example, evidence collected from a crime scene, such as a bottle and glass, may contain touch DNA from individuals who handled those items. The principle of "every contact leaves a trace," which is articulated by Edmond Locard, indicates the importance of physical evidence at a crime scene relevant to DNA examination (Alketbi, 2023). It also happens while collecting presumptive evidence and during testing periods. The presumptive process must measure that width and length for a court document report. This idea underpins contemporary forensic methodologies, such as DNA analysis, by mandating that investigators seek and interpret residual evidence to recreate events and identify perpetrators. The geometric characteristics of that evidence, the size of objects, or the volume of evidence are often critical for analysis, identification, and linkage. The current process involves obtaining these measurements through direct contact tools such as calipers, rulers, and tapes. In the current method the investigator will take photographs, including the measurement tool for reference. Although these methods are well-established in practice, they are susceptible to inter-observer variability and parallax error, as well as providing substantial risk of contaminating (Wyatt, 2014; Balk, 2015; Forensic Science Regulator, 2023, 2024) or destroying vulnerable evidence (Van Gastel *et al.*, 2021), especially for touch DNA on evidence.

Two-dimensional photogrammetry uses a camera with a reference scale (Cheng *et al.*, 2022). The accuracy of this method depends on keeping the camera perfectly perpendicular and ensuring that the evidence and the reference scale are in the same plane. Standardizing this process is difficult given the variability of evidence. Operational efficiency is further limited by the type and abundance of target objects.

Several significant examples of the use of this technology include criminal investigation applications (Sansoni *et al.*, 2009). While photogrammetry offers high-resolution texture mapping, it inherently suffers from scale ambiguity and fails in textureless or non-Lambertian regions (Seitz *et al.*, 2006; Szeliski, 2011). In contrast, depth camera technology, or Time-of-Flight (ToF) cameras (Hansard *et al.*, 2012; Park & Yi, 2024), suggest a new way of carrying out forensic domain metrology. The active depth sensors provide an

absolute metric scale and robust performance in low-light (Khoshelham & Oude Elberink, 2012) and low-texture environments (Henry *et al.*, 2012). It can reduce time-consuming presumptive periods and avoid evidence contamination.

The main characteristic of depth cameras is that sensors acquire a comprehensive point cloud of the surrounding environment. While functioning similarly to conventional two-dimensional cameras, depth sensors are distinct in their ability to record depth information for each individual pixel (Bordbar *et al.*, 2019).

To minimize procedural duration and mitigate the risk of evidence contamination, this study develops and validates an automated system employing a depth camera for the non-contact assessment of forensic evidence (Balk, 2015; Wyatt, 2014). A primary objective is to demonstrate that this 3D methodology (Bordbar *et al.*, 2019; Carew *et al.*, 2021; Sheshtar *et al.*, 2025) overcomes the inherent limitations of 2D photogrammetry by acquiring precise, volumetric data from point clouds analogous to applications (Liu *et al.*, 2025). This experiment can apply the technique to the forensic science domain. This approach allows a rapid, streamlined presumptive forensic inquiry that preserves the chain of custody while ensuring seamless data integration with DNA laboratories (Badiye *et al.*, 2023; Law 101: Legal Guide for the Forensic Expert. Chain of Custody, 2023). Although we are using a depth camera in this work, the document report to court still accepts 2D images. Therefore, the other advantage of a depth camera is efficient measurement in presumptive work. We also bind automatic measurement with a touchless technique by applying previous work on Integration of Object Detection in Crime Scene Investigation (Werukanjana *et al.*, 2025) by upgrading to a depth camera with YOLO object detection to the presumptive process. These are all contributions between the computer science and forensic science domains by utilization techniques. The efforts are touchless and less time-consuming when compared with manual and 2D photographic methods.

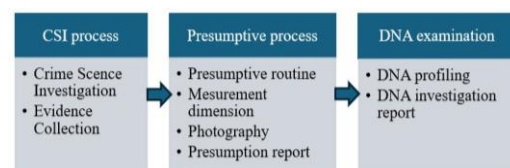


Figure 1. The presumptive process in DNA laboratory

Materials and Methods

Task on presumptive forensic DNA laboratory

In this section, we do not conduct an exhaustive examination of the methodologies employed in the acquisition of presumptive evidence. Instead, we focus on a particular category of evidence collection packages and evidence inside that are forwarded for forensic analysis. Figure 1 illustrates the preliminary biological screening procedures. Before transmitting these sampling traces to the DNA analytical apparatus, we evaluate the presumptive evidence contained within them by assessing factors such as the dimensions of the evidence sample, integrity, and potential contamination (Balk, 2015; Forensic Science Regulator, 2023, 2024). Based on the image provided, this diagram outlines a three-step process for forensic investigation related to evidence that involves DNA. Photography and measurement of physical evidence are essential steps in the preliminary examination. As illustrated in Figure 2, the manual photography method used in the presumptive phase is a time-consuming step with a high risk of contamination.



Figure 2. The manual method on the presumptive process that risks contamination and is time consuming

Presumptive process

The second stage involves preliminary testing and documentation of the evidence, which includes a presumptive routine: a quick initial test to determine if a substance is likely to be blood, semen, or another biological fluid. If the evidence is physical, the measurement dimensions and photography steps are used to document the location and state of the evidence before it is sent for further analysis. A presumption report has been created to summarize these preliminary findings.

DNA examination

In the final phase, comprehensive laboratory examinations are conducted with precision and accuracy. The collected evidentiary material undergoes DNA profiling, an intricate process that

aims to discern a distinct genetic profile from the gathered specimens. This generated profile may be compared with established samples or pre-existing databases. The final stage is compiling a DNA examination report, which effectively contains the analytical results that require applicable precision and legal proceedings.

Data Acquisition

We reference a set of evidence samples of 6 categories with manual measurements in Table 1. We can see that the evidence varies in length because in criminal cases the evidence will not be complete, for example some tools or equipment that can be weapons in crime. In life we transform all evidence into average values for evaluation in this study.

This dataset is a collection of records acquired from manual measurements in a presumptive DNA laboratory where objects are measured and timed during a specific operation. The data provides details on six distinct types of evidence samples (Bottle, Glass, Knife, Scissors, Gun, and Tool). While the table suggests there are 30 records for each type (e.g., 'BOTTLE-001' to 'BOTTLE-030'), the sample points provided allow for an analysis of the structure and the general range of values. Based on the small sample of data points provided (the first three and the 30th entry for each type), we can observe differences across object types in their physical characteristics and the time required for their operation.

Table 1. Evidence Samples

Sample ID	Type	Length (cm)	Width (cm)	Time (Sec)
BOTTLE-001	Bottle	15.20	6.80	45.00
BOTTLE-002	Bottle	22.50	8.10	38.00
BOTTLE-030	Bottle	10.80	5.40	60.00
GLASS-001	Glass	9.50	7.20	48.00
GLASS-002	Glass	12.80	8.50	35.00
GLASS-030	Glass	13.70	8.90	30.00
KNIFE-001	Knife	18.00	3.50	49.00
KNIFE-002	Knife	25.50	4.20	32.00
KNIFE-030	Knife	26.00	4.30	35.00
SCISSORS-001	Scissors	12.50	7.50	48.00
SCISSORS-002	Scissors	20.00	9.00	38.00
SCISSORS-030	Scissors	16.70	8.00	31.00
GUN-001	Gun	18.20	13.50	49.00
GUN-002	Gun	25.00	15.00	38.00
GUN-030	Gun	26.50	15.10	30.00
TOOL-001	Tool	14.50	5.00	49.00
TOOL-002	Tool	28.00	7.50	38.00
TOOL-030	Tool	21.50	6.30	35.00

We employed a commercially available Time-of-Flight (ToF) depth camera that can measure with

sub-millimeter accuracy (Hraud *et al.*, 2016; Xu *et al.*, 2022). The apparatus was mounted on a movable support structure, such as a tripod or stand, and the top-view depth camera was positioned approximately 50 cm above the forensic evidence during this experiment. The calibration process measures the object to its precise dimensions of 5 cm × 5 cm × 5 cm. This configuration enables the eradication of occlusions and the maintenance of a consistent view. DNA analysis was conducted on forensic samples, including various liquid spatter patterns with established volumetric capabilities, tool impressions, as well as general and high-frequency applications from criminal scenes. The camera accurately monitored millions of X, Y, and Z coordinates to generate a three-dimensional point cloud for each forensic sample.

However, forensic evidence is frequently recovered in a fragmented or deformed state (e.g., a damaged bottle found at a crime scene), rendering standard comparisons against intact reference objects ineffective.

We executed a comprehensive analysis of two distinct modalities of efficiency: temporal efficiency and accuracy, utilizing a wide array of evidence types commonly encountered at crime scenes such as bottles, glass fragments, knives, scissors, firearms, and various tools; with each category comprising thirty samples. The assessment of temporal efficiency involves measuring the interval from the initiation of the measurement process until the evidence, along with its dimensional data, is totally archived in a repository, quantified in seconds. We evaluated the accuracy of our approach in relation to conventional manual techniques and a basic 2D camera system, employing a depth camera for measurement within the identical temporal parameters. Ultimately, we scrutinized accuracy by investigating errors and relative errors across all samples in juxtaposition to manual methodologies.

Automated Measurement techniques

This experiment combines object detection by YOLO of previous work and presumptive measurement. However, we upgraded by utilizing a depth camera with the advantage of its properties. These techniques assist the processing of point cloud data. A solution was developed, which used the Open 3D module in Python programming. The framework for establishing an object's dimensions includes several rigorous processes that result in a 3D point cloud with geometric assessments. The major goal is to first obtain the object's three-dimensional coordinates and then use proven methods to calculate its dimension. This creates an

application with the main goal assessing the effectiveness of time-consuming procedure. A minor goal is touchless evidence (Ramsey, 2022; Termanini, 2023) for preventing contamination during the evidence presumptive phase. The application and components are shown in Figure 3.

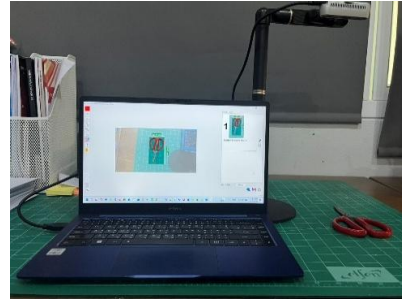


Figure 3. Application and components for presumptive with depth camera

Point Cloud Generation

In the presumptive process, we use evidence examples. They can be weaponized or have some touch DNA on them. First, we need to formalize the conversion of the depth map from the camera into a 3D point cloud. This is based on the pinhole camera model, which relates 2D pixel coordinates to 3D world coordinates, as shown in Figure 4.

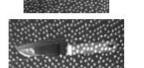
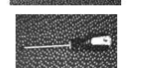
Evidence	RGB	Cloud Point
Bottle		
Glass		
Knife		
Scissors		
Gun		
Tool		

Figure 4. Evidence Sample of RGB and Point cloud

Figure 5 visualizes the software pipeline used in Open3D to convert Time-of-Flight (ToF) data into a 3D point cloud. The technique composes the following main steps.

- Input Acquisition (Scene & Depth Camera)

- Calibration (The 'Key' to 3D)
- Formulation (The 2D to 3D Conversion)
- Result Visualized (3D Point Cloud)

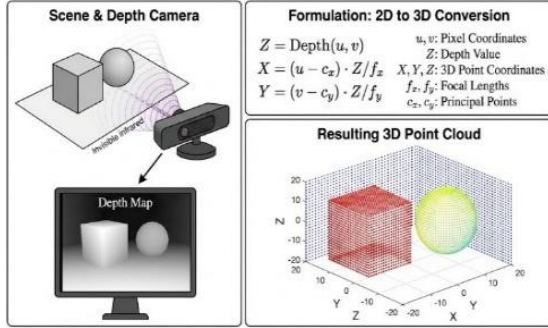


Figure 5. Depth Camera workflow outline and point cloud

Point Cloud Representation

A point cloud is a set of data points in a 3D coordinate system. Each point P_i is typically represented by (x, y, z) coordinates in a global or camera-centric reference frame:

$$P = \{P_1, P_2, P_3, \dots, P_N\}$$

Where:

$$P_i = \{x_i, y_i, z_i\} \in R^3$$

General Measurement Formulation (Euclidean Distance)

If we identify two points, top right P_A and bottom left P_B that correspond to the extremes of the dimensions that we want to measure (e.g., the scissors and end of the handle found in a crime scene), size D is the Euclidean distance between them.

$$D = \|P_A - P_B\|^2$$

$$= \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2}$$

Where:

- $P_A = (x_A, y_A, z_A)$ are the coordinates of the first point.
- $P_B = (x_B, y_B, z_B)$ are the coordinates of the second point.
- D is the measured distance (size)

Evidence Size Measurement with a Depth Camera

To measure the size of evidence, using the example of a pair of scissors, the general workflow

involves image capture, point cloud generation, segmentation, key point identification, and distance calculation.

Step 1: Point Cloud Generation from Depth Camera

A depth camera (like an RGB-D camera) provides an RGB image and a corresponding depth map. The depth map gives the distance (d) from the camera to the object for each pixel (u, v) . The 3D coordinates (x, y, z) for each pixel point can be calculated using the camera's intrinsic parameters (focal lengths f_x, f_y and principal point c_x, c_y).

Formulation for Point Generation:

$$z = d$$

$$x = (u - c_x) * \frac{z}{f_x}$$

$$y = (v - c_y) * \frac{z}{f_y}$$

Where:

- (u, v) is the pixel coordinate in the image.
- d is the depth value at (u, v) .
- (x, y, z) are the resulting 3D coordinates in the camera's frame.
- f_x, f_y, c_x, c_y are the calibrated intrinsic camera parameters.

Step 2: Evidence Measurement Process

- Object Segmentation: The point cloud must first be segmented to isolate only those points belonging to the scissors from the background and other objects (e.g., using color, depth, or geometric features). Let the segmented scissor point cloud be $P\{evidence\}$.
- Key point Identification (Length): To measure the length of the scissors, we need to find the two points that are farthest apart. This is often done by calculating the bounding box (like an oriented bounding box, OBB) that tightly encloses $P\{evidence\}$. The length is then the largest dimension of this bounding box.
- Key point Identification (Width): For width or thickness, the OBB's second and third dimensions provide these measurements. Alternatively, we might need to identify specific features by width.
- Distance Calculation: Once the two key points P_{tip}, P_{handle} (or the dimensions of the OBB) are identified, the actual measurement is calculated using the

Euclidean Distance Formula (D) from the general formulation.

With the length of the evidence, a common approach is to find the maximum distance between any two points in the segmented point cloud $P\{evidence\}$.

$$Length = \frac{MAX}{P_A, P_B \in P\{evidence\}} \left(\sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2} \right)$$

Depth cameras typically capture spatial data in three dimensions, with coordinates calculated in centimeters based on the object's dimensions along the x, y, and z axes. However, for DNA laboratory examination reports, only the length and width are required. The distances were computed along the x and y axes, as illustrated in Figure 6.

This method enables automated measurement

Evidence Type	Length(cm)	Width(cm)
	30.09	3.00
	23.53	4.45
	10.05	10.02
	20.06	3.01
	23.50	14.30
	17.50	7.56

Figure 6. The application and sample of evidence measurement with a depth camera

without requiring the use of several cameras or the manual placement of reference objects. The advantages include reducing time during preliminary measurements and avoiding contamination of the evidence.

Validation Study

The experiment gathered 180 samples of bottles, glasses, knives, scissors, firearms, and tools. The accuracy of the depth camera system was validated against ground truth data. The physical dimensions of the evidence were precisely

measured using a laser scanner. The study is measured and evaluated by comparing between a manual and simple camera below.

- The automated depth camera system.
- Automated simple camera system.
- Manual measurements taken by forensic technicians using digital calipers.

Statistical analysis, including mean absolute error and standard deviation, was used to evaluate the accuracy and precision of each method.

Results and Discussion

The validation study confirmed that the depth camera-based system provided the most time-saving and precise measurements. The automated 3D system records the time of processing, starting from the acquired images until they are saved to the repository, ready for the report. While using this 3D system, we recorded a percent relative error of 180 samples. The experimental results are summarized in Table 2. The result shows a mean absolute error of 3D in length and width of 0.23 with a standard deviation of 0.16. In contrast, the manual measurements produced a mean error of 2D photogrammetry, yielding a mean error in length and width of 1.51 and 1.39, respectively. The larger standard deviation in the traditional methods highlights their susceptibility to human and systematic errors. The increased standard deviations in older approaches underscore their vulnerability to human and systematic errors.

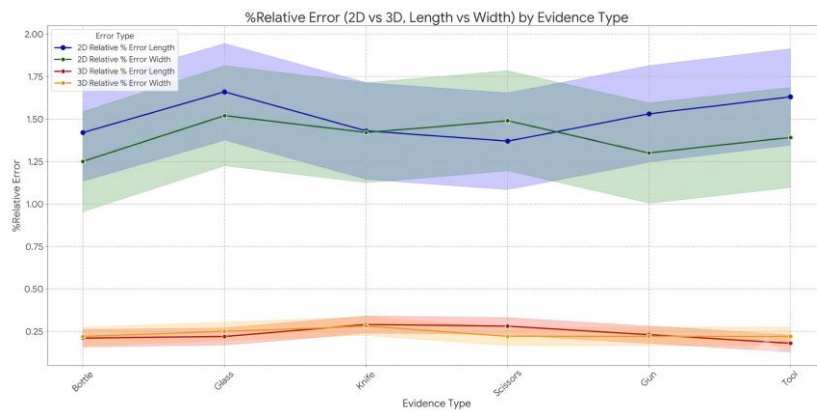
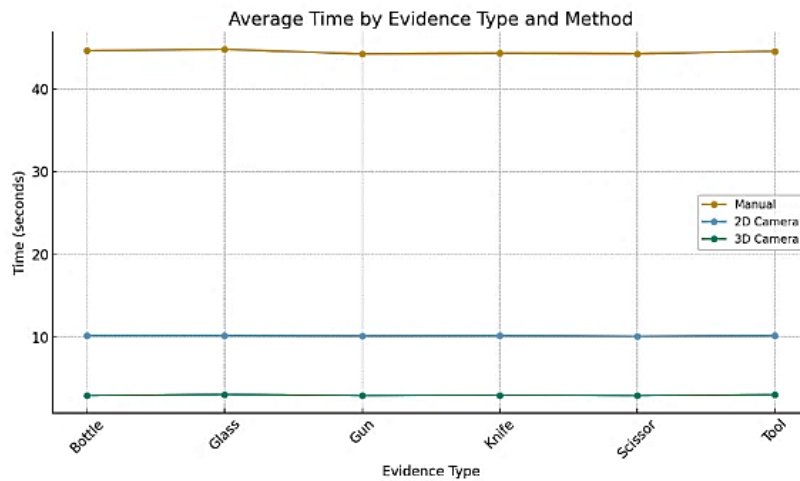
Figure 7 illustrates the percentage relative error across six types of evidence (bottle, glass, knife, scissors, gun, and tool). The results indicate that 3D modeling is substantially more accurate than 2D modeling, exhibiting both fewer relative errors and greater consistency (lower variance) in length and width measurements.

Comparison of Depth Camera Time

Figure 8 illustrates the differences between manual measurement time and depth camera processing time across 180 samples. The orange line represents the Manual Time (sec), and the green line represents the Depth Camera Time (sec). The manual time is significantly higher than the depth camera time for all samples. The manual time fluctuates between approximately 44 and 45 seconds, while the depth camera time is consistently much lower, with two measurement methods ranging from 5 to 11 seconds. This chart clearly illustrates the substantial differences, ranging from 5 to 11 seconds.

Table 2. Comparative Analysis of Measurement

Sample	Manual Time (sec)	Simple Camera Time (sec)	Depth Camera Time (sec)	%Relative Error 2D		%Relative Error 3D	
				Length (cm)	Width (cm)	Length (cm)	Width (cm)
Bottle	44.63	10.20	2.39	1.42	1.25	0.21	0.22
Glass	44.80	10.20	3.07	1.66	1.52	0.22	0.25
Knife	44.33	10.17	2.97	1.43	1.42	0.29	0.28
Scissors	44.27	10.10	2.93	1.37	1.49	0.28	0.22
Gun	44.23	10.13	2.93	1.53	1.30	0.23	0.22
Tool	44.57	10.17	3.03	1.63	1.39	0.18	0.22
Average	44.47	10.16	2.88	1.51	1.39	0.23	0.23
STD	8.74	2.81	1.35	0.85	0.88	0.15	0.16

**Figure 7. %Relative Error by Evidence Type****Figure 8. Comparison of Manual, 2D and 3D Depth Camera operation time**

A comparison of average recording times for different evidence categories using manual, 2D, and 3D camera methods is presented in Figure 8. The inclusion of 180 samples from six distinct groups of evidence contributed to an enhancement in the statistical robustness of the validation study. Across

all categories, the depth camera system consistently delivered the lowest standard deviations and sub-millimeter mean errors, demonstrating an exceptional level of precision and consistency. With 2D camera measurement, the percentage of relative errors was higher than the 3D depth camera when

compared with the manual technique. The demonstration is particularly helpful for various sorts of physical evidence: the bigger sample set demonstrated that the automated 3D depth camera technique can fulfill its promise for unique presumptive investigations.

In general, the chart provides confirmation that both two-dimensional and three-dimensional procedures yield accurate results, with extremely small errors for both length and width, depending on the type and state of evidence. The chart compares % relative error with inspection of Error Band across evidence types. Both Length and Width errors stay within a narrow band (~1.2%–1.6%). There is no strong evidence that measuring length or width produces higher errors. All evidence types show stable error rates, without big fluctuations. This means measurement reliability is consistent across different evidence categories.

The results unequivocally demonstrate the advantages of using depth cameras for forensic metrology. The system's ability to capture a complete 3D representation of an object solves the core problems of perspective distortion and planar displacement that affect 2D photogrammetry.

The primary innovation of this study is a precise method for measuring length, width, and depth. This is especially significant for forms of evidence such as bite marks, imprints, and fluid patterns, where the three-dimensional aspect holds vital information. The automated analytical pipeline eliminates observer subjectivity, yielding an objective and reproducible record suitable for courtroom presentation and inter-laboratory validation. Although the existing method is exceptionally efficient, it may be compromised by highly reflective or clear surfaces such as bottles or glass that can disrupt the ToF sensor. In the future, research will investigate sensor fusion, integrating depth data with high-resolution color images and polarized light filters to address these material constraints because the nature of evidence is malformed objects (Kuriyal, 2025).

Instrumentation and Software Environment

The primary data acquisition device employed in this study was the Intel Real Sense Depth Camera D435i. This active stereo infrared camera features an integrated Inertial Measurement Unit (IMU), which provides six-degrees-of-freedom (6DoF) motion data essential for stabilizing the handheld measurement protocols used in this research.

For software implementation, the device was interfaced using the Intel RealSense SDK 2.0. All subsequent data processing, including point cloud generation and manipulation, was developed using

Python scripts integrated with the Open3D library for 3D data visualization and analysis.

Conclusions

This study demonstrates that Time-of-Flight (ToF) depth cameras outperform conventional 2D imaging and manual measurement methods in presumptive forensic DNA laboratories. The proposed methodology provides a rapid, non-contact, and highly accurate approach for creating comprehensive 3D documentation of physical evidence. By capturing objective and complete geometric data directly at crime scenes, ToF technology substantially improves the precision of early-stage evidence documentation. As such, ToF devices represent an important advance in modern forensic investigation. Future research should focus on expanding the size and diversity of forensic evidence datasets. Large-scale and diverse datasets encompassing various types of evidence will enhance the development and validation of machine learning algorithms, resulting in higher accuracy. Datasets that cover diverse environmental conditions such as crime scenes under different lighting conditions, various surface materials, and different types of physical evidence will enable the creation of robust models applicable to real-world field scenarios.

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Author Contributions

Conceptualization: P.W.; Methodology: P.W.; Investigation: P.W., S.M., K.C., N.P.; Data Curation: N.P.; Writing - Original Draft: P.W.; Writing - Review & Editing: P.W., S.M., K.C., N.P., P.S.; Supervision: P.S. All authors have read and agreed to the published version of the manuscript.

Ethics Statement

This study was reviewed and officially approved by the Human Research Ethics Committee of the Faculty of Forensic Science, Royal Police Cadet

Academy. (Certificate of Approval No. 680610-003 and date of approval: June 10, 2025)

Declaration of Generative AI Use

During the preparation of this manuscript, the authors used Gemini Pro to generate portraits instead of real human photographs to ensure strict compliance with the Personal Data Protection Act (PDPA) and Quillbot (Learneo) to assist with grammar correction, copyediting, and improving the overall readability of the text. All AI-assisted in preliminary screening by summarizing English literature to speed up the identification of relevant studies and helped refine academic tone and ensure consistent use of terminology. All AI-assisted outputs were critically reviewed, verified, and revised by the authors, who accept full responsibility for the content of this manuscript

Conflict of Interest

The authors declare that they have no conflicts of interest related to this work.

Data Availability Statement

The data used in this study are confidential and not publicly available due to legal and privacy restrictions associated with forensic casework evidence.

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